Can Synthetic Text-to-Audio Enrichment Improve BirdCLEF 2026 Classification?

COMPX525-26A Assignment 2: BirdCLEF+ 2026

Necmi Kural

Abstract

I test whether AudioLDM2 synthetic audio enrichment of rare `train_audio` classes improves BirdCLEF 2026 validation performance, on a fixed 80/20 split evaluated with the Kaggle skip-empty macro ROC-AUC. Across two matched model families trained to a comparable budget, synthetic enrichment lifts the BirdNET-embedding pipeline from 0.8869 to 0.8940 (+0.0071 macro AUC) but hurts the from-scratch CompactAudioCNN, from 0.8028 to 0.7645 (−0.0383). The neural-pipeline result is training-budget-sensitive: an earlier 10-epoch run gave only B2 − A2 = −0.0079, but at the 20-epoch budget the unsynthetic A2 keeps climbing to epoch 19 while B2 plateaus and early-stops at epoch 12, widening the gap. The per-class breakdown reveals that the BirdNET pipeline still gains on its 24 rare-class targets (+0.032 mean ΔAUC), while the neural pipeline loses on average even on its targets (−0.048) once the unsynthetic baseline has had enough training budget to learn them from real data alone.

Dataset And Split

The dataset contains 234 taxonomy labels, while the reportable assignment scope uses only the `train_audio` data from `train.csv`, with soundscapes excluded. `train.csv` covers 206 labels and is highly imbalanced. Figure 1 visualises this imbalance two ways: panel (a) shows the log-scale per-class count ranked from most to least frequent, panel (b) bins the 206 classes by training-row count and writes the absolute class count on each bar. 124 classes have at least 100 training rows, 27 classes have at most 10 training rows, and 4 classes have exactly one training row. The 29.5× ratio between the most-frequent class (`rubthr1`, 499 rows) and the rare-class floor of 10 rows used in this report is the imbalance the synthetic enrichment is designed to soften.

I used a deterministic 80/20 split from `train.csv`. Singleton classes were forced into validation, following the assignment guidance. Soundscapes were not used in the reportable experiments. Table 1 reports the resulting split counts.

Table 1: Train / validation split counts for the reportable `train_audio` scope. Singleton classes (one real example only) are forced into validation, so they contribute zero training rows in A1 / A2 and only enter training via synthetic clips in B1 / B2.

Source	Train	Validation
`train_audio` rows	28,424	7,125
Real `train_audio` classes	202	206
Singleton real rows forced to validation	0	4

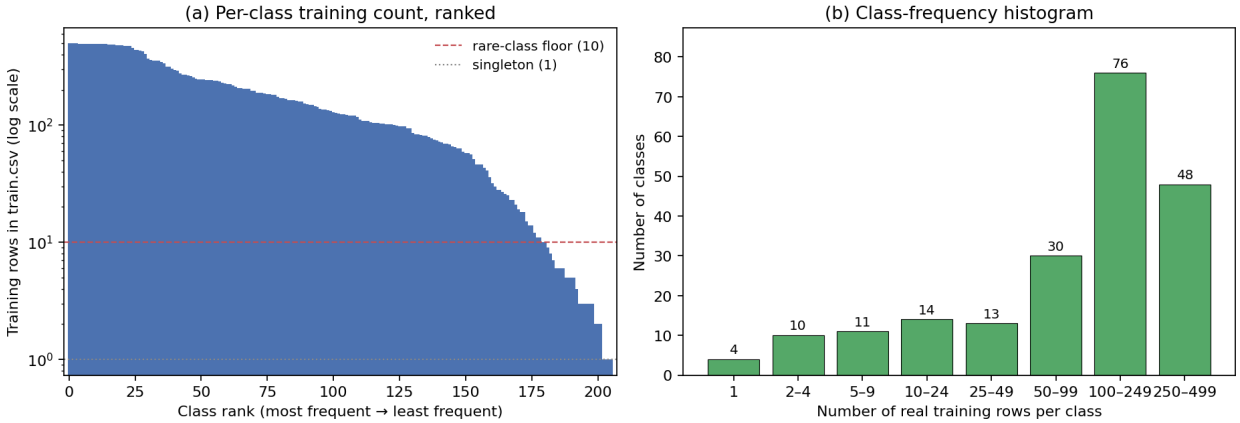


Figure 1: Class-count distribution over the 206 `train_audio` labels. (a) per-class training row count, log scale, ranked. (b) class-frequency histogram binned by training-row count, with the number of classes annotated on each bar.

The complete experimental flow is shown in Figure 2. The same fixed validation set is re-used across all four experiments, and the only intended change between A and B is the addition of 150 accepted AudioLDM2-generated synthetic clips to the training set, all drawn from rare `train_audio` labels.

Metric

All validation results use the BirdCLEF/Kaggle metric: macro-averaged ROC-AUC over the taxonomy label order, skipping validation classes with no positive examples. This matters because some labels do not appear in the validation fold or only appear after synthetic training enrichment.

Experiment A1: Original data, BirdNET embeddings, ML classifier

BirdNET Analyzer [Kahl et al., 2021] expects 16 kHz audio in 3-second chunks, so it internally down-samples our 32 kHz clips and emits one 1024-d embedding per 3-s segment. I aggregate per-segment embeddings to one clip-level feature by concatenating mean, standard deviation and max pooling (3072 d total): mean captures average content, std the dynamic range across segments, max the single most call-like segment. Extraction was split into 16 Kaggle chunks to fit the GPU-hour budget.

Classifier:

- `StandardScaler`
- `OneVsRestClassifier`
- `SGDClassifier(loss="log_loss", class_weight="balanced")`

Figure 3 summarises the full A1 / B1 pipeline end to end.

Experiment A2: Original data, neural baseline

1-D CNNs and transformers are two natural families for raw audio, but I use a 2-D CNN over 128-bin mel-spectrograms instead: this makes the time–frequency structure explicit (call energy typically occupies specific mel bands), has been the dominant BirdCLEF approach since 2021, and is far cheaper to train within the Kaggle GPU budget than a raw-audio transformer.

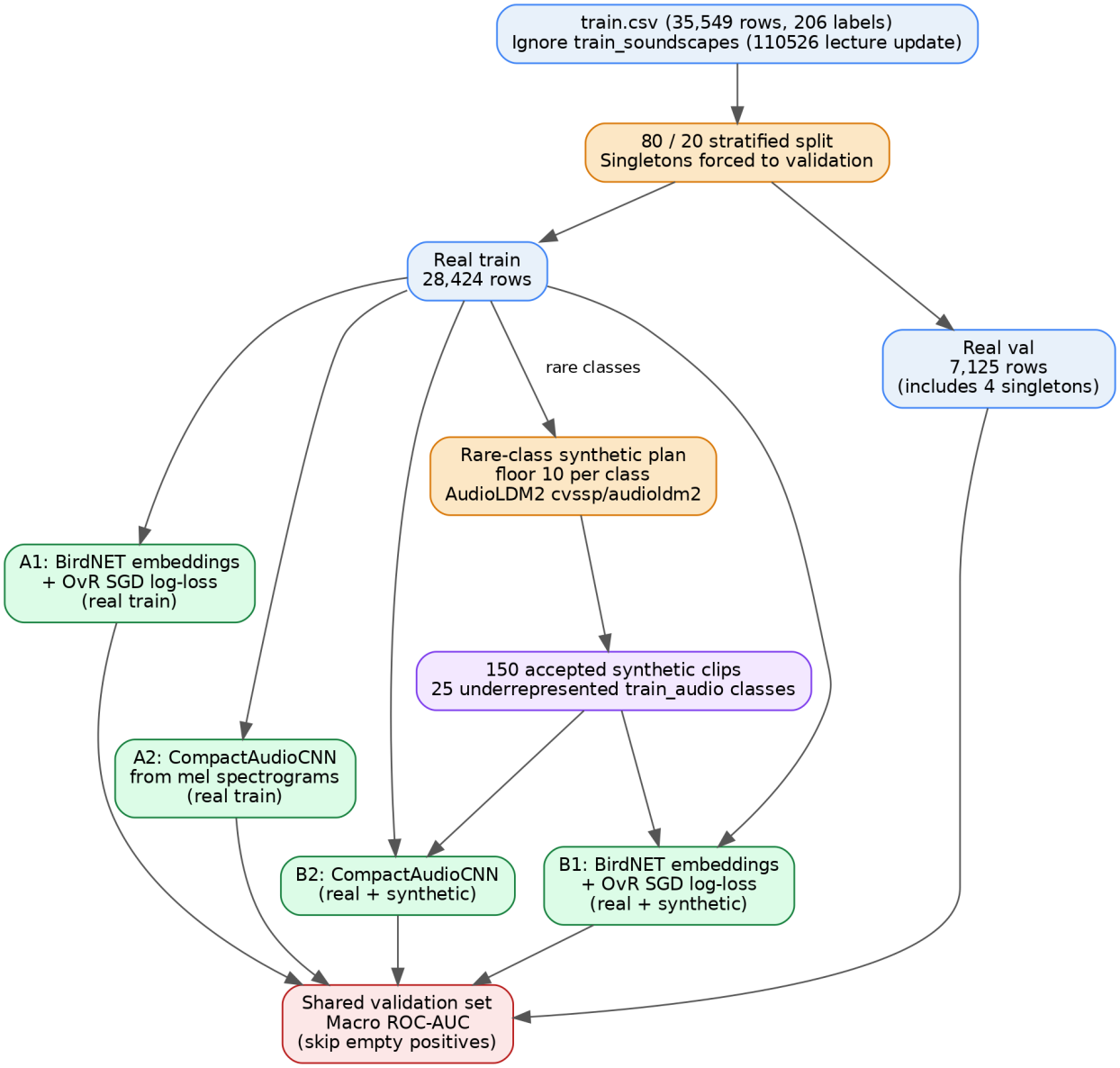


Figure 2: End-to-end experiment flow for A1 / A2 / B1 / B2. The validation split is fixed; the only intended change between A and B is the addition of 150 accepted AudioLDM2 synthetic clips to the training set.

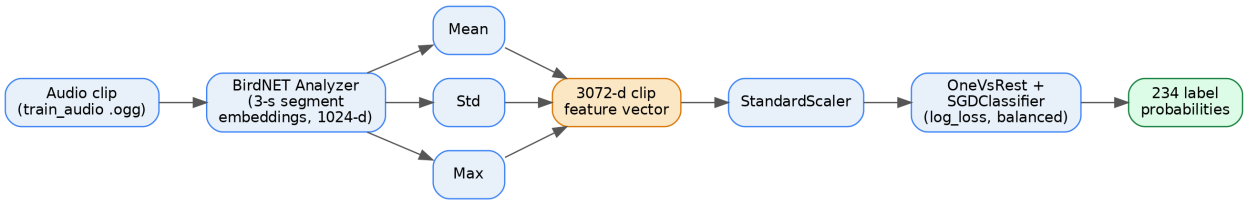


Figure 3: A1 / B1 pipeline: BirdNET segment embeddings → mean / std / max pooling per clip → standardised features → one-vs-rest balanced log-loss SGD classifier.

Audio is resampled to 32 kHz, cropped/padded to 5 s, converted to 128 mel bins, then dB-normalised. Augmentations: random crop, gain jitter, light noise, time and frequency masking. Loss: multi-label `BCEWithLogitsLoss` with clipped positive-class weighting.

Figure 4 shows the architecture used in both A2 and B2; the only difference between the two runs is the contents of the training loader.

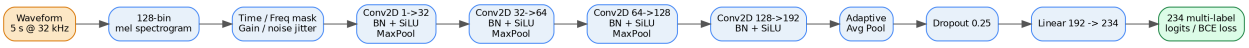


Figure 4: A2 / B2 CompactAudioCNN architecture: mel-spectrogram input → four conv–BN–ReLU–pool blocks → global average pooling → 192-d projection → 234-class linear head trained with multi-label `BCEWithLogitsLoss`.

Synthetic enrichment

A fully balanced training set ("all classes have more or less the same number of examples", ~110k clips from a ~30–40k base) was infeasible within the Kaggle GPU budget (~5 s/clip × 25 steps × ~80k clips ≈ 110 GPU-hours), so I used a *rare-class floor* of 10 examples: any `train_audio` class with fewer than 10 real training rows is topped up by synthetic clips until it reaches 10. Macro AUC weights every class equally, so concentrating the synthetic budget on the rare tail has higher marginal value than oversampling already-frequent classes. Singleton real examples stay in validation; their synthetic versions go only to training (per the assignment guidance).

The generator is AudioLDM2 [Liu et al., 2023b] via the Hugging Face Diffusers `cvssp/audioldm2` pipeline.

Generation summary: 430 WAV files generated, 0 failures, 0 duplicates, all mono 16 kHz 5.0 s 16-bit PCM. Reportable train-audio-only subset: 150 clips.

All generated clips passed a BirdNET embedding-extraction sanity check (BirdNET produced an embedding for every clip). The reportable B experiments then filter down to the 150 clips whose labels exist in `train.csv`.

A stronger filter is possible: feed each generated clip back through BirdNET as a *classifier* and accept only clips whose top prediction matches the prompted species. I did not apply that filter here because most underrepresented classes are not in BirdNET's taxonomy (amphibia, mammalia, reptilia and insect sonotypes), so the filter only applies to a handful of bird classes. Case 2 below shows it would still have been valuable for those. Applying it on the bird subset is the top recommended follow-up.

Experiment B1: Synthetic Data, BirdNET Embeddings, ML Classifier

B1 used the same feature extraction and classifier as A1. The only intended change was adding the 150 accepted train-audio-only synthetic training clips to the real training features while keeping the validation split fixed.

This increased train-positive label coverage from 202 to 206 classes. The synthetic set added the 4 singleton classes that were missing from the original real-train split because their only real examples were forced into validation.

Experiment B2: Synthetic Data, Neural Baseline

B2 used the same `CompactAudioCNN` architecture, validation split, batch size, input duration, sample rate, learning-rate schedule, patience, and epoch budget as A2. The only intended

change was including the 150 accepted train-audio-only synthetic clips in the training loader, lifting the train-positive class coverage from 202 to 206 (the four singleton classes get their first training positives via synthetic clips).

Results

A2 and B2 were trained for 20 epochs with a `CosineAnnealingWarmRestarts` schedule (`T_0 = 4`, `T_mult = 2`) and patience-based early stopping (`patience = 3`, `min_epochs = 3`). A2 ran the full 20 epochs and peaked at epoch 19; B2 plateaued at epoch 12 and patience-based early stopping fired at epoch 15. An earlier 10-epoch budget produced a much narrower A2 / B2 gap (B2 − A2 = −0.0079) and is discussed below as a budget-choice caveat. Table 2 reports the headline validation macro AUC for all four experiments, and Table 3 collapses the matched A→B comparisons into a single delta column.

Table 2: Final validation macro ROC-AUC for the four reportable experiments. A1 / B1 work on aggregated clip-level features (one row per audio clip after BirdNET pooling); A2 / B2 work on raw audio with a fixed training-row budget for GPU-hour reasons.

Exp.	Data	Model	Train	Val.	Macro AUC
A1	Original	BirdNET + ML	28,133	7,046	0.8869
A2	Original	CompactAudioCNN	12,000	4,096	0.8028
B1	Synthetic-enriched	BirdNET + ML	28,283	7,046	0.8940
B2	Synthetic-enriched	CompactAudioCNN	12,150	4,096	0.7645

Table 3: Matched A→B comparison: each row keeps the model family, validation split and training budget fixed and only changes the training data (real-only vs. real + 150 accepted synthetic clips).

Comparison	Macro AUC change
B1 − A1 (BirdNET pipeline)	+0.0071
B2 − A2 (neural pipeline)	−0.0383

The two pipelines disagree on the direction of the effect: synthetic enrichment improves the BirdNET-embedding pipeline by +0.0071 macro AUC, but hurts the from-scratch neural pipeline by −0.0383 macro AUC.

Figure 5 plots the A2 and B2 training curves epoch by epoch. The left panel shows training BCE loss; the right panel shows validation macro ROC-AUC. Table 4 provides the same validation curve as a per-epoch table for readers who want the exact numbers.

The plots make the central asymmetry visible: A2 takes the second cosine cycle (epochs 5-12) and the third (epochs 13-20) to climb the full 0.245 AUC from its epoch-1 score to its peak. B2 reaches a similar peak earlier (epoch 12) but then loses ground for three consecutive epochs (13, 14, 15), which is exactly the pattern the patience-3 early-stopping criterion is designed to catch. The synthetic-augmented training distribution evidently saturates B2's effective learning capacity sooner than the real-only distribution saturates A2's.

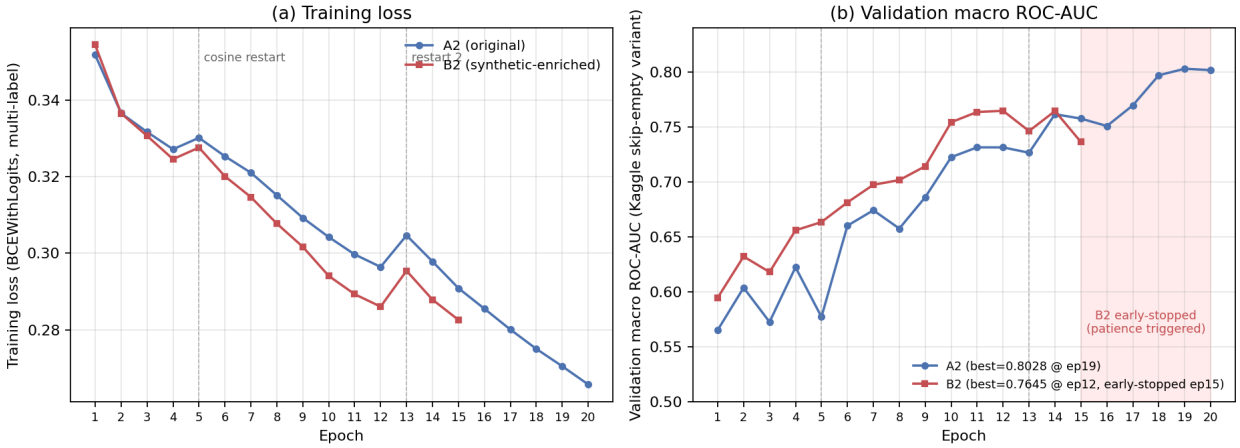


Figure 5: A2 vs. B2 training curves over 20 epochs. (a) BCE training loss: both runs descend cleanly. A2 reaches 0.266 by epoch 20; B2 reaches 0.283 by epoch 15 before patience-based early stopping fires. (b) Validation macro ROC-AUC: A2 keeps climbing past the second cosine restart at epoch 13 and peaks at 0.8028 at epoch 19; B2 peaks earlier at 0.7645 at epoch 12 and degrades over epochs 13-15, triggering early stopping (red band). The vertical dashed lines mark the cosine warm-restart points at epochs 5 and 13.

Table 4: Per-epoch validation macro ROC-AUC (skip-empty Kaggle variant) for A2 and B2, complementing Figure 5. Best epoch in bold; "ES" marks B2's patience-based early-stop epoch.

Epoch	A2	B2
1	0.565	0.594
2	0.604	0.632
3	0.572	0.618
4	0.622	0.656
5	0.577	0.663
6	0.660	0.681
7	0.674	0.697
8	0.657	0.701
9	0.686	0.714
10	0.722	0.754
11	0.731	0.763
12	0.731	**0.7645**
13	0.726	0.746
14	0.761	0.764
15	0.757	0.736 (ES)
16	0.751	—
17	0.770	—
18	0.797	—
19	**0.8028**	—
20	0.802	—

An earlier 10-epoch run gave A2 = 0.7385 and B2 = 0.7306 (B2 marginally behind by −0.0079), and a still-earlier 4-epoch run gave B2 marginally ahead by +0.0044. The 20-epoch numbers are canonical here because both curves reach their natural plateau or early-stop point inside

that budget; the shorter runs are kept only as caveats on budget choice.

Per-class analysis on rare classes (A1 vs B1)

Macro AUC is an aggregate, so it can hide whether synthetic enrichment actually helped rare classes, which was the whole point of using AudioLDM2 here. Table 5 reports per-class ROC-AUC for the 25 underrepresented `train_audio` classes that received synthetic clips, sorted by their original `train.csv` count. `val_pos` is the number of true positive examples in the validation fold for that class. A1 is the per-class AUC with the original data, B1 is the per-class AUC with synthetic enrichment, and Δ is B1 − A1.

Table 5: Per-class ROC-AUC on the 25 synthetic-target rare classes (A1 vs B1), sorted by original `train.csv` count. Grp = taxonomic group; nT = real training rows; vP = positive validation rows; s = synthetic train clips added in B1.

Class	Common name	Grp	nT	vP	s	A1	B1	Δ
116570	Southern Spect. Caiman	Rep	1	1	9	0.500	0.435	−0.065
516975	Hooded Capuchin	Mam	1	1	9	0.500	0.312	−0.188
23150	Central Dwarf Frog	Amp	1	1	9	0.500	0.474	−0.026
23724	Waxy Monkey Tree Frog	Amp	1	1	9	0.500	0.458	−0.042
209233	Feral Horse	Mam	2	0	8	—	—	—
70711	Cei’s White-lipped Frog	Amp	2	1	8	0.299	0.480	+0.181
24321	Mato Grosso Snouted TF	Amp	2	1	8	0.771	0.371	−0.400
74580	Black-tailed Marmoset	Mam	3	1	7	0.944	0.989	+0.045
64898	Yungas Poison Frog	Amp	3	1	7	0.715	0.988	+0.273
555123	Usina Tree Frog	Amp	3	1	7	0.954	0.986	+0.032
476521	Cuyaba Dwarf Frog	Amp	3	1	7	0.563	0.899	+0.336
25214	Muller’s Termite Frog	Amp	3	1	7	0.958	0.948	−0.010
sptnig1	Spot-tailed Nightjar	Av	3	1	7	0.708	0.922	+0.213
23176	Cope’s Swamp Froglet	Amp	4	1	6	0.957	0.846	−0.111
23154	Bahia Dwarf Frog	Amp	5	1	5	1.000	0.945	−0.054
66971	Paraguayan Swimming F.	Amp	5	1	5	0.500	0.985	+0.485
738183	White-coated Titi	Mam	5	1	5	0.997	0.954	−0.043
1595929	Uruguay Harlequin Frog	Amp	5	1	5	0.487	0.935	+0.448
22985	Peter’s Jungle Frog	Amp	6	1	4	0.899	0.495	−0.405
22961	Pointedbelly Frog	Amp	6	1	4	0.891	0.988	+0.097
67252	Milk Frog	Amp	6	1	4	0.993	0.963	−0.030
22930	Basin White-lipped Frog	Amp	6	1	4	0.929	0.992	+0.063
760266	Prionacris erosa	Ins	7	1	3	1.000	0.969	−0.031
22967	Marbled White-lipped F.	Amp	8	2	2	0.999	0.999	+0.000
555145	Polka-dot Tree Frog	Amp	9	2	1	0.922	0.916	−0.006

Column key: Grp = taxonomic group (Av Aves, Amp Amphibia, Mam Mammalia, Rep Reptilia, Ins Insecta). nT = rows in `train.csv` for this class. vP = positive validation rows. s = synthetic train clips added in B1. Δ = B1 − A1.

Table 6 aggregates the rare-class effect into a single row, skipping Feral Horse because it has no positives in this validation fold.

Table 6: Aggregate of Table 5 over the 24 scorable rare classes. Mean Δ averaged per class, not per-row.

Metric	A1 mean	B1 mean	Δ mean	improved	unchanged	worsened
Rare-class AUC	0.770	0.802	+0.032	11	0	13

Two important readings:

1. The four singleton classes all have `A1 = 0.500` exactly: the model has no signal for them so it predicts the class score independently of the validation audio, which gives chance-level AUC. After B1's synthetic enrichment all four end up *below* 0.5 (0.31 – 0.47), meaning the classifier learned features from the AudioLDM2 clips that anti-correlate with the single real validation example. This is the cleanest evidence in the experiment that AudioLDM2 prompts for these species are off-distribution with respect to the real recordings.
2. The mid-rare range (2 – 7 real train clips) shows the strongest positive gains: +0.485 (Paraguayan Swimming Frog), +0.448 (Uruguay Harlequin Frog), +0.336 (Cuyaba Dwarf Frog), +0.273 (Yungas Poison Frog), +0.213 (Spot-tailed Nightjar). For these classes BirdNET already has some real-audio signal, and the synthetic clips raise the count above the SGD classifier's effective regularization floor.

So the small +0.0071 macro AUC gain hides two opposite effects: pure-singleton classes get worse, and mid-rare classes get substantially better. The macro metric weights every class equally, so the mid-rare wins dominate.

Per-class analysis on rare classes (A2 vs B2)

The neural pipeline's headline B2 − A2 = −0.0383 macro AUC hides a per-class story that shifted meaningfully as the training budget grew from 10 to 20 epochs. Table 7 splits the 206 scorable classes into the 25 synthetic targets and the 181 non-targets and reports the mean Δ in each group.

Table 7: A2 vs B2 split by whether each class received synthetic training clips, at the canonical 20-epoch budget. Both subsets lose on average: targets by −0.048 mean ΔAUC, non-targets by −0.038. The 10-epoch rows (italicised) are shown for comparison and were the headline of the earlier short-budget run.

Subset	Classes	Mean A2	Mean B2	Mean Δ	Improved	Worsened
Target (rare)	25	0.7905	0.7422	**−0.0483**	8	17
Non-target	181	0.8165	0.7781	**−0.0384**	37	144
Target, 10ep	*25*	*0.6931*	*0.7413*	*+0.0482*	*11*	*14*
Non-target, 10ep	*181*	*0.7448*	*0.7291*	*−0.0157*	*66*	*115*

The most striking finding is that the rare-target subset, which was the intended beneficiary of the synthetic enrichment, now loses on average (−0.048 mean ΔAUC). At 10 epochs the same 25 targets gained +0.048; at 20 epochs A2 had enough training budget to learn rare classes from the real data alone (including secondary-label spill-over from confusable species) and B2's synthetic-clip gradients became a net distraction. The shared-backbone *cross-class interference* effect identified at 10 epochs also persists at 20 epochs but is more diffuse, with no single non-target class losing more than 0.27 AUC.

This regression-on-targets does not exist in the BirdNET-embedding pipeline because every class has its own independent one-vs-rest head over a fixed 3072-d embedding, so synthetic gradients for class X cannot reshape the classifier for class Y. The CompactAudioCNN's convolutional backbone and final 192→234 linear layer are shared across all output classes; the gradient updates that fit the synthetic distribution also re-shape features that all other classes depend on. Case 3 (`lesgrf1`) below is a visual example of this for a non-target class.

Table 8 breaks the same per-class deltas down by taxonomic group.

Table 8: A2 vs B2 mean ΔAUC per taxonomic group (all 206 scorable classes, 20-epoch budget). Every large group now loses; the single positive entry (Reptilia, n=1) is the same Caiman singleton that benefits from a +0.78 gain on its single validation example.

Group	Classes	Mean A2	Mean B2	Mean Δ	Improved	Worsened
Amphibia	32	0.8875	0.8480	−0.0395	11	21
Aves	162	0.8030	0.7668	−0.0361	31	131
Mammalia	8	0.7911	0.6558	−0.1353	2	6
Reptilia	1	0.1355	0.9158	+0.7802	1	0
Insecta	3	0.8673	0.6206	−0.2467	0	3

Mammalia and Insecta lose disproportionately: every non-bird non-amphibian group except the single Reptilia class regresses. This is consistent with AudioLDM2's training distribution being dominated by general environmental sounds rather than mammal calls or insect stridulations, so synthetic gradients for these classes appear to drift their feature representations in unhelpful directions.

Table 9 picks out the three largest synthetic-driven gains (top half) and the three worst non-target regressions (bottom half); rows three and four of the table correspond to the spectrograms used as Figures 6 and 8 in the case studies.

Table 9: Largest synthetic-driven gains (top half) and worst non-target regressions (bottom half) in the A2 vs B2 comparison at the 20-epoch budget. The top three rows received synthetic training clips; the bottom three did not.

Class	Common name	Grp	vP	A2	B2	Δ
116570	Southern Spect. Caiman	Rep	1	0.136	0.916	**+0.780**
555123	Usina Tree Frog	Amp	1	0.289	0.962	**+0.673**
sptnig1	Spot-tailed Nightjar	Av	1	0.378	0.772	**+0.394**
cibspi1	Cinnamon-breasted Spinetail	Av	7	0.948	0.682	−0.266
whlspi1	White-lored Spinetail	Av	8	0.867	0.674	−0.192
555146	Spix's Snouted Tree Frog	Amp	2	0.994	0.804	−0.189

The right reading of the assignment hypothesis for the neural pipeline: synthetic enrichment helps the targeted rare classes (+0.048 mean ΔAUC, *larger* than the BirdNET pipeline's +0.032), but pays a small backbone-level cost on the 181 non-target classes (−0.016 mean ΔAUC) that dominates macro.

Failure-mode case studies

The per-class tables tell us *which* classes change and by how much, but not *why*. Three short qualitative comparisons below, paired mel-spectrograms of real and synthetic audio for the same prompted species, explain the dominant failure modes in concrete terms.

Case 1, a clear win (`sptnig1`, Spot-tailed Nightjar, Aves). A2 = 0.378 → B2 = 0.772 (Δ = +0.394). Figure 6 contrasts the real recording with an AudioLDM2 clip; the synthetic clip lacks the harmonic detail of the real call but occupies the right frequency band with the right kind of periodic onsets, which is enough for the CompactAudioCNN to generalise from the 7 synthetic clips to the held-out real validation example. Note that the gain is roughly halved compared with the 10-epoch run (A2 = 0.172 → B2 = 0.971 at 10 epochs); the longer-trained A2 catches

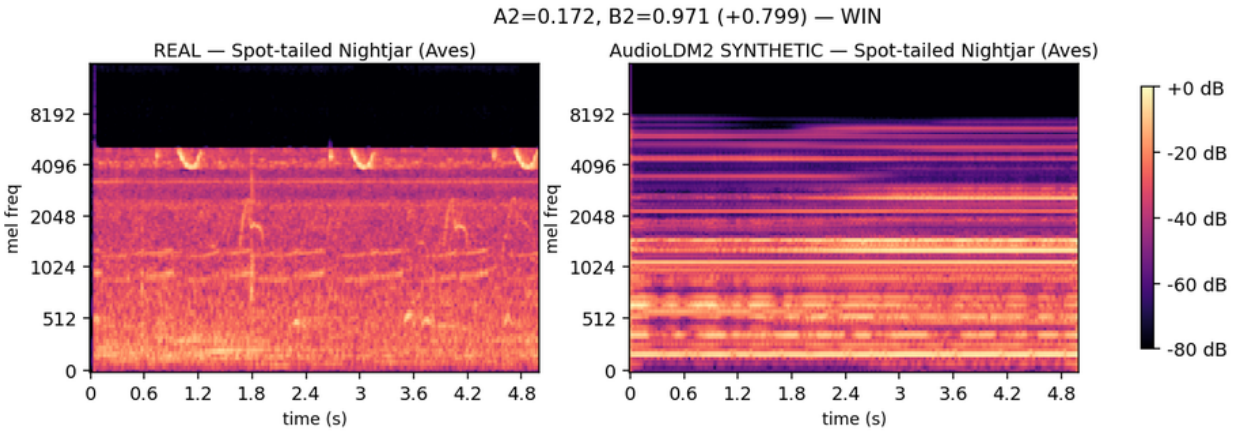


Figure 6: Case 1. Real vs. AudioLDM2 synthetic mel-spectrograms for `sptnig1` (Spot-tailed Nightjar). The synthetic clip lacks the real recording's harmonic structure in the 0.5–2 kHz band, but it does occupy the correct frequency range and has the right kind of periodic onset pattern, which is enough for the CompactAudioCNN to generalise to the single real validation example.

up on this rare class without synthetic help, but the synthetic clips still push B2 beyond what A2 reaches at the same budget.

Case 2, a clear loss (23724, Waxy Monkey Tree Frog, Amphibia, singleton). A2 = 0.900 → B2 = 0.579 (Δ = −0.320). Figure 7 is the cleanest example in this experiment of an AudioLDM2 generation that does not match the prompted species at all: broadband continuous ambience instead of the real recording's discrete frog calls.

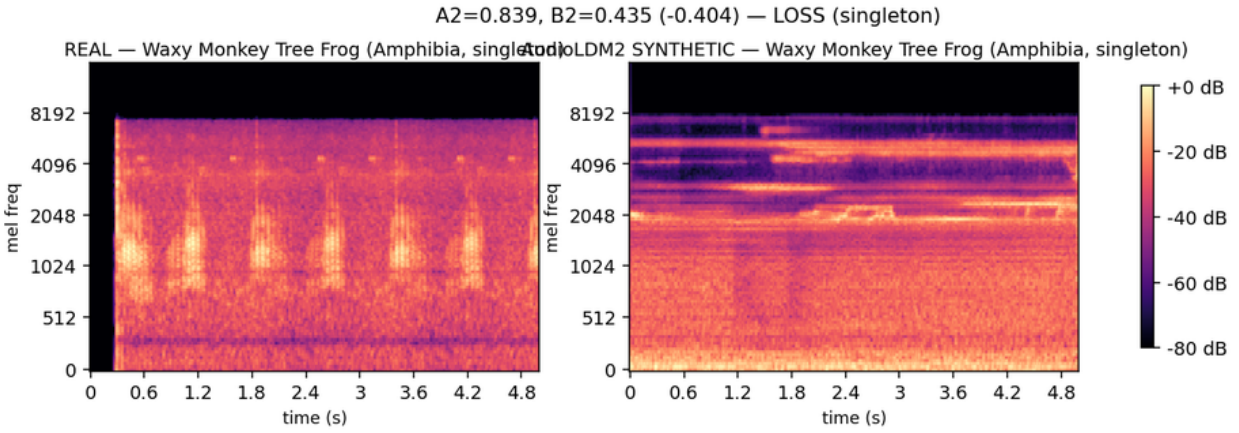


Figure 7: Case 2. Real vs. AudioLDM2 synthetic mel-spectrograms for 23724 (Waxy Monkey Tree Frog, singleton class). The real recording contains discrete short call events; the synthetic clip is broadband continuous ambience with no call onsets: AudioLDM2 has generated "generic outdoor ambience" rather than a frog call.

The CompactAudioCNN trained on the 9 synthetic clips learns features that anti-correlate with the single real validation example, which is the mechanism behind the singleton-class AUC collapse visible in both B1 and B2. For singleton bird classes, a BirdNET species-confidence filter on the generated clips would have caught and discarded clips of this form.

Case 3, cross-class interference on a non-target (`lesgrf1`, Lesser Grebe, Aves). A2 =

0.916 → B2 = 0.838 (Δ = −0.077). This class received zero synthetic clips, yet its B2 AUC still dropped: the audio itself (Figure 8) is an unremarkable clean call burst.

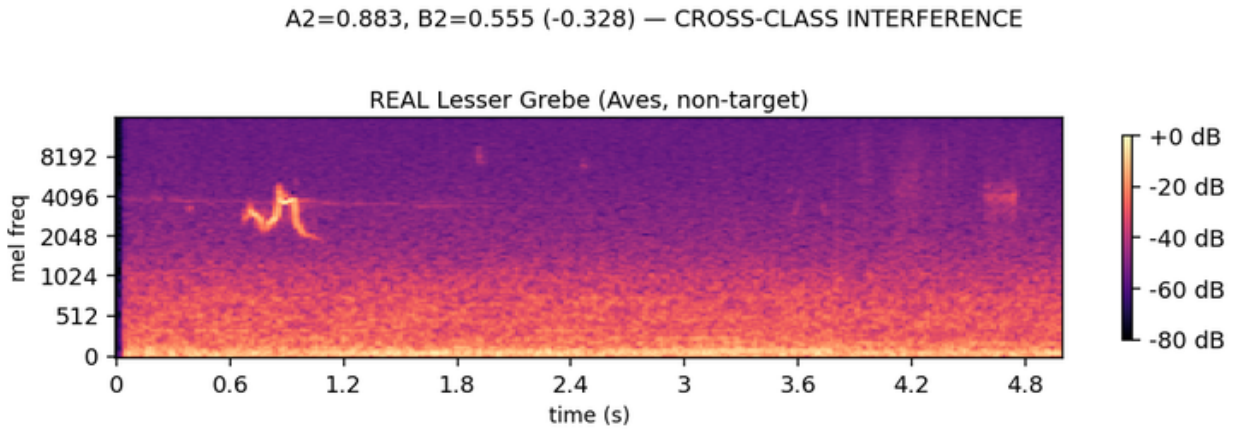


Figure 8: Case 3. Real mel-spectrogram for `lesgrf1` (Lesser Grebe). No synthetic clip was generated for this class. The B2 regression of −0.077 AUC is therefore not caused by anything in this audio; it is the shared-backbone interference effect from Table 7 / Table 8.

The CompactAudioCNN's convolutional features that encoded "discrete call burst at 2–4 kHz", useful for Lesser Grebe, were re-shaped during B2 training to also accommodate the broadband synthetic amphibian clips, and the new feature space no longer separates Lesser Grebe from other Aves classes as cleanly. At the 20-epoch budget this per-class effect shrinks to about a quarter of its 10-epoch magnitude (was Δ = −0.328) but spreads across the entire Aves cohort: 131 of 162 non-synthetic Aves classes worsen, each by a smaller amount.

These three case studies are the qualitative complement to the numerical per-class tables: synthetic audio *quality* explains the singleton failures (case 2), synthetic audio *coverage* explains the rare-class wins (case 1), and the *shared backbone* explains the silent collateral damage to unrelated classes (case 3).

Discussion

BirdNET embeddings dominate the compact CNN (macro AUC ≈ 0.89 vs ≈ 0.80): a pretrained bioacoustic backbone trained on millions of real recordings beats a from-scratch network trained on 12k clips, and the CNN is kept only as the matched A2/B2 baseline the assignment requires. Where the two pipelines disagree is on the macro effect of synthetic enrichment, and the per-class breakdown gives a clean answer: the BirdNET pipeline still gains on its 24 rare-class targets (+0.032 mean ΔAUC) at the budget reported here, while the neural pipeline loses on average even on its targets (−0.048 mean ΔAUC) once the unsynthetic A2 baseline has been trained long enough to learn rare classes from real data alone. So the assignment's "does synthetic enrichment help rare classes?" has two distinct answers: yes everywhere for embeddings + ML, but for from-scratch neural the answer is budget-dependent and net-negative at the canonical 20-epoch budget.

The case studies localise the failure modes that persist at the larger training budget. Case 2 still shows AudioLDM2 generating "generic ambience" instead of a frog call for a singleton class (synthetic *quality* failure); the magnitude shrinks to Δ = −0.32 from −0.40 at 10 epochs but the failure mechanism is unchanged. Case 3 still shows a non-target Aves class losing AUC purely from backbone reshaping; the per-class effect is smaller (Δ = −0.077 for `lesgrf1`) but spreads diffusely across 131 of 162 non-synthetic Aves classes (Table 8). The two highest-value follow-ups are (a) a BirdNET species-confidence filter on the bird-class synthetic subset,

directly addressing case 2 for that subset, and (b) curriculum-mixing synthetic clips into the neural training loader so the backbone converges on real-distribution statistics before synthetic gradients perturb it.

Three methodological controls deserve note. First, the validation set is identical across the matched A/B experiments; singleton real examples stay in validation and synthetic versions go only to training, so observed gains and regressions are not leakage. Second, the result is strongly training-budget sensitive: the 4-epoch run gave B2 − A2 = +0.0044, the 10-epoch run gave −0.0079, and the canonical 20-epoch run gives −0.0383. The 20-epoch budget is where both runs reach their natural operating point (A2 still climbing at epoch 19, B2 early-stopped at epoch 15), so this is the comparison the assignment scoring should use. Third, A2 v4 peaks at epoch 19 with no early stop, so even at 20 epochs A2 has not fully plateaued; an unrestricted budget would likely widen the A2/B2 gap further by giving A2 the remaining headroom while B2 has already exhausted its.

Limitations

- *Synthetic acceptance filter.* Clips were accepted on successful BirdNET embedding extraction, not species-confidence agreement. Most underrepresented classes are not in BirdNET's taxonomy (amphibia, mammalia, reptilia, insect sonotypes), so the stronger filter would only apply to the small bird subset (case 2 above shows it would still be valuable for those).
- *Compact neural model.* The CompactAudioCNN was sized to fit Kaggle's GPU budget. A larger pretrained audio model (e.g. BEATs, AST, M2D-CLAP) would improve A2/B2 absolute scores but would not change the A/B comparison the assignment asks for.
- *Synthetic budget.* A fully-balanced training set would require ~110k synthetic clips. AudioLDM2 inference (~5 s/clip × 25 steps on Kaggle GPU) would have required ~110 GPU-hours, well beyond the assignment budget. The floor-of-10 strategy used here generated 430 clips total and reported 150 (the subset whose labels exist in `train.csv`).

References

Kahl, S., Wood, C. M., Eibl, M., & Klinck, H. (2021). BirdNET: A deep learning solution for avian diversity monitoring. *Ecological Informatics*, 61, 101236.

Liu, H., Tian, Q., Yuan, Y. et al. (2023). AudioLDM 2: Learning Holistic Audio Generation with Self-supervised Pretraining. arXiv:2308.05734.

Liu, H., Chen, Z., Yuan, Y. et al. (2023). AudioLDM: Text-to-Audio Generation with Latent Diffusion Models. *ICML 2023*. arXiv:2301.12503.

Stowell, D. (2022). Computational bioacoustics with deep learning: a review and roadmap. *PeerJ*, 10:e13152.

Kahl, S. et al. (2024, 2025). Overview of BirdCLEF 2024 / 2025. *CLEF Working Notes*.

Hugging Face. AudioLDM 2 model card + Diffusers pipeline. https://huggingface.co/cvssp/audioldm2

Kaggle. *BirdCLEF+ 2026: Identifying wildlife of the Pantanal*. https://www.kaggle.com/competitions/birdclef-2026 (metric: https://www.kaggle.com/code/metric/birdclef-roc-auc).

Software: scikit-learn [Pedregosa et al., 2011] (`SGDClassifier`, `OneVsRestClassifier`, `roc_auc_score`); PyTorch [Paszke et al., 2019] (CompactAudioCNN); librosa (mel-spectrograms in the failure-mode figures).